

# Workflow: Describing a point pattern

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## DEFINE THE EVENTS

**`st_nearest_feature()` `st_distance(by_element = TRUE)`**

a nearest-neighbor distance of zero marks records that share a location; remove those that are not events

then

## DEFINE THE WINDOW

**`st_buffer()` `st_filter()`**

the area in which an event could have been recorded; buffer the mapped boundary where the search passed it, and state the choice, because every statistic depends on it

then

## INTENSITY

**`st_intersects() %>% lengths() / st_area()`**

events per unit of area: how many, not how arranged

and

## ARRANGEMENT

**`st_make_grid()` `var(nests) / mean(nests)`**

quadrat counts at the scale of the quadrat: the ratio is above one for a clustered pattern; nearest-neighbor distances describe the spacing without a grid

then

## COMPARE WITH CHANCE

**`st_sample()` `inside map_dbl()`**

random patterns of the same size in the same window; report where the observed statistic falls among them, with the seed and the number of draws